Session 5: Decision Rules - Critical Value vs. p-Value

Context & Objective

Textbook: Chapter 9 – Hypothesis Testing & Decision Rules

Objective: Students implement decision rules, calculating test statistics and comparing p-values to alpha to approve or reject process changes

1. Antigravity Skill Definition (hypothesis-test-decision-maker)

Skill Name: hypothesis-test-decision-maker

Purpose: Takes a test statistic and computes the p-value. Then applies the p-value rule and the critical-value rule separately. If both methods agree, it outputs a reject-or-fail-to-reject decision in words a manager can act on.

1.1 Core Logic & Formulas

Step 1: Gather the Inputs

- Test statistic value (z or t)
- Is it a right-tailed, left-tailed, or two-tailed test?
- Significance level (α)
- Critical value (z_crit or t_crit)

Step 2: Calculate the p-Value

Right-tailed: p-value = $P(Z \geq z_{calculated})$

Left-tailed: p-value = $P(Z \leq z_{calculated})$

Two-tailed: p-value = $2 \times P(Z \geq |z_{calculated}|)$

Step 3: Method 1 - Use the p-Value

If p-value $\leq \alpha$: Reject H_0

If p-value $> \alpha$: Fail to Reject H_0

Step 4: Method 2 - Use the Critical Value

If |test statistic| ≥ |critical value|: Reject H_0

If |test statistic| < |critical value|: Fail to Reject H_0

Step 5: Confirm Both Methods Agree

The two methods reach the same decision every time. Once you verify they match, write a conclusion a manager can understand and act on.

2. Student Execution Prompt

Evaluate a hypothesis test for delivery times.
- Test type: Upper-tailed.
- Significance level (alpha) = 0.05 (critical z = 1.645).
- Calculated z-statistic = 1.88.

Compute the p-value, apply the decision rules, and output the reject or accept decision

Given Data & Test Setup:

Test Type	Right-tailed (upper-tailed)
H_0: Null Hypothesis	Average delivery time = 24 hours
Ha: Alternative Hypothesis	Average delivery time > 24 hours
Calculated z-statistic	1.88
Significance level (α)	0.05 (5%)
Critical z-value (z_crit)	1.645

Assignment Task:

1. Calculate the p-value for z = 1.88 on a right-tailed test.
2. Apply the p-value rule: is p-value ≤ 0.05?
3. Apply the critical-value rule: is |z| ≥ 1.645?
4. Confirm both rules reach the same decision.
5. Final Verdict: reject or fail to reject H_0, and why.

3. Output & Interpretation

The Numbers:

z-statistic	1.88
p-value	0.0301 (3.01%)
Significance level (α)	0.05 (5.00%)
Critical value (z_crit)	1.645

Method 1: p-Value Rule

p-value (3.01%) < α (5.00%) → Reject H_0

Method 2: Critical Value Rule

z-statistic (1.88) > z_crit (1.645) → Reject H_0

Decision: Both methods agree. Reject H_0

What It Means:
The p-value of 0.0301 says this: if delivery times really averaged 24 hours, there is only a 3% chance we would see a test statistic as extreme as 1.88. That is low enough that it makes sense to reject the idea that times are still at 24 hours. The data point toward longer delivery times.

This is not a borderline result. The z-statistic sits comfortably past the cutoff, and the p-value is less than half of alpha. Delays are real.

What the Manager Does:
The operations manager terminates the courier contract. The test proves the delays are not random. The firm begins talks with alternative logistics providers. A formal exit notice goes to the current courier service.

4. Inference

- The p-value method and the critical-value method give the same answer. They are two sides of the same coin.
- A p-value is the chance of getting a result this far from H_0 if H_0 is actually true. It is not the chance that H_0 is true.
- When $p \leq \alpha$, reject H_0. When $p > \alpha$, fail to reject H_0. Do not accept H_0—just acknowledge a lack of evidence against it.
- Small p-value plus large practical difference equals a strong business case. Small p-value alone is not enough.
- Misinterpreting p-values is rampant. A small p means your data are unlikely if H_0 is true, not that your effect is important.

